Organic Chemistry

1. But-2-ene exhibits cis-trans-isomerism due to

   A. rotation around C_2-C_3 double bond
   B. rotation around C_3-C_4 sigma bond
   C. rotation around C_1-C_2 bond
   D. restricted rotation around C=C bond

2. What is the general formula for cycloalkanes?

   A. C_nH_{2n+2}
   B. C_nH_{2n}
   C. C_nH_{2n-2}
   D. CnH_{2n+1}

3. Rotation of plane polarized light can be measured by:
   A. manometer
   B. calorimeter
   C. polarimeter
   D. viscometer

4. The compound having molecular formula $C_4H_{10}O$ can show-
   A. Metamerism
   B. Functional isomerism
   C. Positional isomerism
   D. All of the above

5. Tautomerism will be exhibited by
   A. $(CH_3)_3CNO$
   B. $(CH_3)_2NH$
   C. R_3CNO_2
   D. RCH_2NO_2

6. A compound of molecular formula C_7H_{16} shows optical isomerism, the compound will be
   A. 2,3-dimethyl pentane
   B. 2,2-dimethyl butane
   C. 2-methyl hexane
   D. None of the above

7. A compound that is formed on passing chlorine through propene at 400°C is -
   A. Vinyl chloride
   B. Allyl chloride
   C. Ethyl chloride
   D. 1,2-Dichloro ethane

8. Hydroxylation of propyne in the presence of $HgSO_4/H_2SO_4$ is initiated by the attack of
   A. carbene
   B. free radical
   C. electrophile
   D. nucleophile

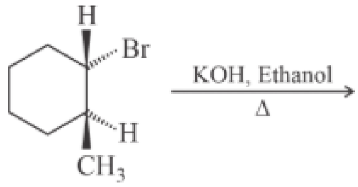
9. Which products are obtained from the ozonolysis of buta-1,3-diene?

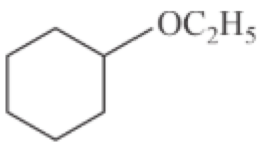
A. Formaldehyde and glyoxal
B. Acetaldehyde and glyoxal
C. Carbon dioxide and glyoxal
D. Formaldehyde, glyoxal, and acetaldehyde

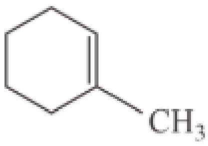
10. When 3,3-dimethyl-2-butanol is heated with conc. H_2SO_4, the major product obtained is -

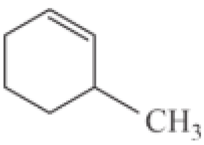
A. 2,3-Dimethyl-2-butene
B. 3,3-Dimethyl-2-butene
C. 2,3-Dimethyl-1-butene
D. None of these

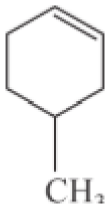
11. What is major product of following reaction?

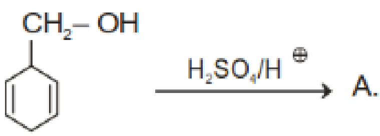


A. 

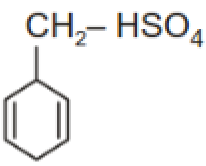
B. 

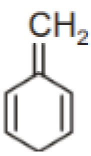
C. 

D. 

12. 

The product A is :

A. 

B. 

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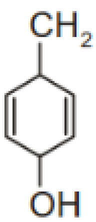
D. 

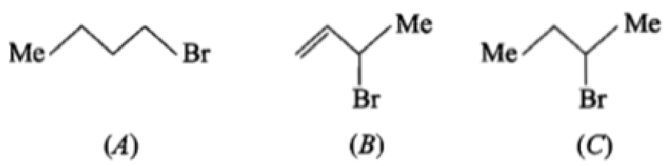
13. The S_N1 mechanism for substitution reaction by nucleophile is favored by:

A. Low concentration of nucleophile
B. Weak nature of the nucleophile
C. Polar solvent
D. All of the above

14. The formation of cyanohydrin from a ketone is an example of:

A. electrophilic addition
B. nucleophilic addition
C. nucleophilic substitution
D. electrophilic substitution

15. Consider the following bromides:



the correct order of S_N1 reactivity is:

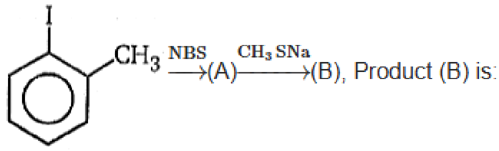
A. A>B>C
B. B>C>A
C. B>A>C
D. C>B>A

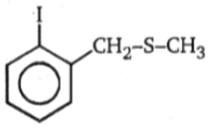
16. Which alcohol is most reactive for dehydrogenation or dehydration?

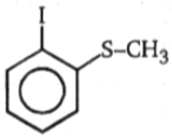
A. CH_3-CH(OH)- CH_2 - NO_2
B. CH_3-CH(OH) - CH_2 - CHO
C. CH_3-CH(OH)- CH_2 - CN
D. CH_3-CH(OH)-CH_2 - $COCH_3$

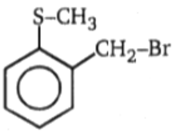
17. Butanenitrile may be prepared by heating-

A. Propyl alcohol with KCN
B. Butyl chloride with KCN
C. Butyl alcohol with KCN
D. Propyl chloride with KCN

18. 

A. 

B. 

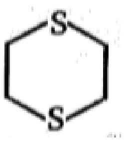
C. 

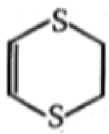
D. None of these

19. Addition of KI accelerates the hydrolysis of primary alkyl halides because:

A. KI is soluble in organic solvents
B. the iodide ion is a weak base and a poor leaving group
C. the iodide ion is a strong base
D. the iodide ion is a powerful nucleophile as well as a good leaving group

20. 1-2-dichloro ethane + $NaSCH_2CH_2SNa \rightarrow P\ (C_4H_8S_2)$
Unknown product (p) of the above reaction is:

A. 

B. 

C. H-S-CH=CH-CH=CH-S-H

D. H-CH=CH-CH_2-S-H

21. Which set of reactants among the following produces anisole?

A. CH_3CHO, RMgX
B. C_6H_5OH, NaOH, CH_3I
C. C_6H_5OH, neutral $FeCl_3$
D. C_6H_5-CH_3, CH_3COCl, $AlCl_3$

22. Propene, CH_3-CH=CH_2 can be converted into 1-propanol by oxidation. Indicate which set of reagents amongst the following is ideal to effect the above conversion?

A. $KMnO_4$(alkaline)
B. Osmium tetroxide (OsO_4/CH_2Cl_2)
C. B_2H_6 and alk H_2O_2
D. O_3/Zn

23. The most effective reagent used to converts but-2-enal to but-2-enol is

A. $KMnO_4$
B. $NaBH_4$
C. H_2/Pt
D. $K_2Cr_2O_7/H_2SO_4$

24. 23g of sodium react with CH_3OH to give

A. 1 mole of O_2
B. 1/2 mole of H_2
C. 1 mole of H_2
D. None of the above

25. Phenol and benzoic acid can be distinguished by:

A. Aqueous $NaHCO_3$
B. Aqueous $NaNO_3$
C. Aqueous NaOH
D. Conc. H_2SO_4

26. The value of C-O-C angle in ether molecule is

A. 180°
B. 150°
C. 90°
D. 110°

27. The products formed in the following reaction, $C_6H_5\text{-}O\text{-}CH_3 + HI \rightarrow$ are

A. C_6H_5OH and CH_3I
B. C_6H_5I and CH_3OH
C. $C_6H_5CH_3$ and HOI
D. C_6H_6 and CH_3OI

28. Primary alcohols can be obtained from the reaction of the RMgX with:

A. HCHO
B. H_2O
C. CO_2
D. CH_3CHO

29. Glycol condenses with ketones to give:

A. Cyclic acetals
B. Cyclic ketals
C. Acetaldehyde
D. oxalic acid

30. Ethyl propanoate on reduction with $LiAlH_4$ yields:

A. Methanol
B. Ethanol and Propanol
C. Propane
D. Mixture of ethanol and methanol

31. The reagent used for the separation of acetaldehyde from acetophenone is:

A. $NaHSO_3$
B. $C_6H_5NHNH_2$
C. NH_2OH
D. NaOH and I_2

32. HCHO and HCOOH are distinguished by treating with:

A. Tollens reagent
B. $NaHCO_3$
C. Fehling's Solution
D. Benedict Solution

33. Which of the following has most acidic proton?

A. CH_3COCH_3
B. $(CH_3)_2C=CH_2$
C. $CH_3COCH_2COCH_3$
D. $(CH_3CO)_3CH$

34. Benzaldehyde and acetaldehyde can be distinguished by:

A. Iodoform test
B. 2,4 - DNP test
C. NH_3 reaction
D. Wolff-Kishner's reduction

35. Fehling's solution is:

A. Acidified copper sulphate solution
B. Ammoniacal cuprous chloride solution
C. Copper sulphate, Rochelle salt + NaOH
D. None of the above

36. Which compounds will not reduce Fehling's solution?
A. Methanal
B. Ethanal
C. Trichloroethanal
D. Benzaldehyde

37. The enolic form of acetone contains:

A. 9 σ-bonds, 1π-bond and 2 lone pairs
B. 8 σ-bonds, 2π-bonds and 2 lone pairs
C. 10 o-bonds, 1π-bond and 1 lone pair
D. 9 σ-bonds, 2π-bonds and 1 lone pair

38. Which one of the following compounds on treatment with $LiAlH_4$ will give a product that will give a positive iodoform test?

A. CH_3CH_2CHO
B. $CH_3CH_2COOCH_3$
C. $CH_3CH_2OCH_2CH_3$
D. CH_3COCH_3

39. Cyanohydrin of which compound gives lactic acid on hydrolysis?

A. Acetone
B. Acetaldehyde
C. Propanal
D. HCHO

40. Ethyl acetate on reaction with excess of methyl magnesium chloride and dil. H_2SO_4 gives

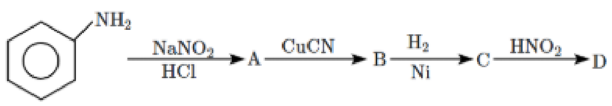
A. dimethyl ketone
B. iso propyl alcohol
C. ethyl aceto acetate
D. t-butyl alcohol

41. $CH_3CH_2Cl \xrightarrow{\text{NaCN}} X \xrightarrow{\text{Ni/H}_2} Y \xrightarrow{\text{Acetic anhydride}} Z$

In the above reaction sequence, Z is

A. $CH_3CH_2CH_2NHCOCH_3$
B. $CH_3CH_2CH_2NH_2$
C. $CH_3CH_2CH_2CONHCH_3$
D. $CH_3CH_2CH_2CONHCOCH_3$

42. Aniline in a set of reactions yield the following products



The structure of the product D would be

A. $C_6H_5CH_2NH_2$
B. $C_6H_5NHCH_2CH_3$
C. C_6H_5NHOH
D. $C_6H_5CH_2OH$

43. What is the end product in the following sequences of operations?

$$\text{Acetamide} \xrightarrow{P_2O_5} A \xrightarrow{4H} B$$

A. CH_3NH_2
B. $C_2H_5NH_2$
C. CH_3CN
D. CH_3COONH_4

44. The least basic aqueous solution among the following is:

A. $(CH_3)_2NH$
B. $CH_3\ NH_2$
C. $(CH_3)_3N$
D. $C_6H_5NH_2$

45. Which one of the following on reduction with $LiAlH_4$ yields a secondary amine ?

A. Methyl isocyanide
B. Acetamide
C. Methyl cyanide
D. Nitroethane

46. Which of the following statements about primary amines is false ?

A. Alkyl amines are stronger bases than aryl amines
B. Alkyl amines react with nitrous acid to produce alcohols
C. Aryl amines react with nitrous acid to produce phenols
D. Alkyl amines are stronger bases than ammonia

47. Among the following, the strongest base is:

A. $C_6H_5NH_2$
B. P-NO_2-$C_6H_4NH_2$
C. m-NO_2-$C_2H_4NH_2$
D. $C_2H_5CH_2NH_2$

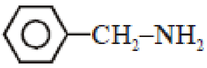
48. Aniline is reacted with bromine water and the resulting product is treated with an aqueous solution of sodium nitrite in presence of dilute hydrochloric acid. The compound so formed is converted into a tertafluoroborate which is subsequently heated. The final product is:

A. 1,3,5-tribromobenzene
B. p-bromofluorobenzene
C. p-bromoaniline
D. 2,4,6-tribromofluorobenzene

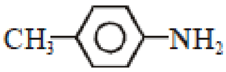
49. $CH_3CH_2NH_2$ contains a basic NH_2 group, but CH_3CONH_2 does not, because;

A. acetamide is amphoteric in character
B. in $CH_3CH_2NH_2$ the electron pair on N-atom is delocalized by resonance
C. in $CH_3CH_2NH_2$ there is no resonance, while in acetamide the lone pair of electron on N-atom is delocalized and therefore less available for protonation
D. none of the above

50. Which of the following cannot be formed by Gabriel phthalimide synthesis:-

A. 

B. $(CH_3)_2CH\text{-}NH_2$

C. 

D. $C_2H_5NH_2$

ANSWER KEY

QN	ANS	QN	ANS	QN	ANS	QN	ANS	QN	ANS
1.	D	11	C	21	B	31	A	41	A
2.	A	12	C	22	C	32	B	42	D
3.	C	13	D	23	B	33	D	43	B
4.	D	14	B	24	B	34	A	44	D
5.	C	15	B	25	A	35	C	45	B
6.	A	16	A	26	D	36	D	46	C
7.	B	17	D	27	A	37	A	47	D
8.	C	18	A	28	A	38	D	48	D
9.	A	19	D	29	B	39	B	49	C
10.	A	20	A	30	B	40	D	50	C